MILITARY SPECIFICATION

DIRECTION FINDER SET AN/ARN-89( )

This amendment forms a part of Military Specification MIL-D-55647(EL), dated 27 August 1970, and is approved for use by all Departments and Agencies of the Department of Defense.

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3.14.1 thru 3.14.2.1: Delete in its entirety and substitute "3.14.1 Temperature-Altitude Comb (see 4.8.1). - The unit(s) shall be subjected to and meet the temperature-altitude requirements with a non-operating minimum exposure of 24 hours at -65° F, atmospheric pressure, and 4 hours minimum exposure at +160° F and atmospheric pressure. The unit(s) shall be operable and conforming to limits specified from -25° F through +160° F. The unit(s) shall be operable from -65° F to -25° F and no measurement shall degrade by more than 6db if specified in db or by more than a factor of 2 if the limit is specified in anything but db, the input voltage shall be from 24 to 29 VDC."

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3.15.2.12.3, Line 2: Delete "0.1 Volt" and substitute "0.01 Volt".

3.15.2.12.3, Line 4: Delete "20 percent" and substitute "26 percent".

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3.15.3.3: Delete "When using the "L-Loop-R" Switch, the speed of the loop rotation shall be continuously variable and have a speed of 25 degrees +5 -0 per second." and substitute "When using the "L-Loop-R" Control, the speed of the loop rotation shall be continuously variable. The speed shall be 25 to 39.7 degrees per second with the "L-Loop-R" Control in either the full "L" or full "R" position."

FSC 5826

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3.15.3.4: Delete in its entirety and substitute "3.15.3.4 Switch, reversing, loop (see 4.9.3.4). - When the loop reversing switch is operated to the top position and the loop mode of operation is selected and the loop L/R switch is operated to position "R", the bearing indicator shall rotate counter-clockwise; and clockwise when the loop L/R switch is operated to position "L". When the loop reversing switch is operated to the bottom position and the loop mode of operation is selected and loop L/R switch is operated to position "R", the bearing indicator shall rotate clockwise; and counter-clockwise when the loop L/R switch is operated to position "L"."

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3.15.4.14: Delete in its entirety and substitute "3.15.4.14 Filtering, sense and loop preamplifier (see 4.9.4.14). - There shall be a minimum attenuation of 26db in the loop preamplifiers and 34db in the sense preamplifiers in the frequency range from 30 to 400 MHz. The phase difference between the loop and sense signals shall not exceed $\pm 33^{\circ}$ over the 100 KHz to 3.0 MHz band."

Under 3.16.2, insert the following:

"3.16.3 Method 3001 and 3002 Conducted Interference (30 Hz to 100 Hz) (see 4.10.3). - The unit(s) shall comply with Method 3001 and 3002, except that Figure 1 limits shall apply.

3.16.4 Method 5005, Rejection of Undesired Signal at Input Terminals (30 Hz to 10 GHz) (see 4.10.4). - The unit(s) shall comply with Method 5005, except that a single signal test may be used and the upper frequency limited to 10 GHz in lieu of 20 GHz specified in MIL-STD-826.

3.16.5 Method 6001, Radiated Susceptibility (14 KHz to 10 GHz) (see 4.10.5). - The unit(s) shall comply with Method 6001, except that: (1) the upper frequency shall be limited to 10 GHz in lieu of 20 GHz specified in MIL-STD-826, (2) the field strength from 1 GHz to 10 GHz shall be limited to 1 volt per meter, (3) the on-course frequency $\pm$50 KHz shall not apply and (4) the test limit of the IF frequency (12.5 MHz) $\pm$50 KHz shall be reduced to 1 volt per meter field strength.

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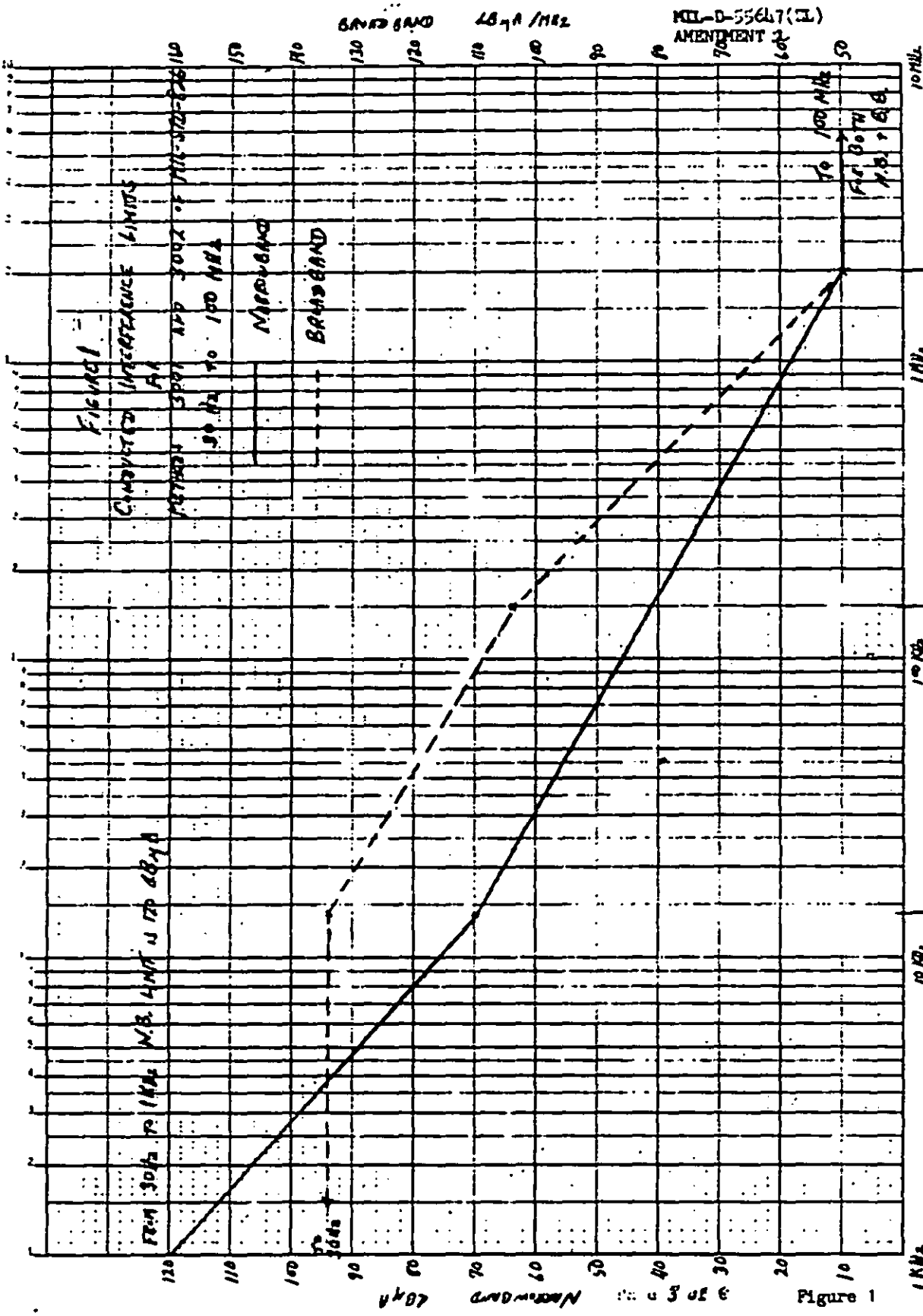


Figure 1

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3.16.6 Method 5002, Conducted Susceptibility (150 KHz to 400 MHz) (see 4.10.6). - The unit(s) shall comply with Method 5002, except that at the on-course frequency, a frequency band of ±50 KHz shall not apply."

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Table III, Group C Inspection: Under 3.16, insert "3.16.1, 3.16.2, 3.16.3, 3.16.4, 3.16.5 and 3.16.6"; under 4.10, insert "4.10.1, 4.10.2, 4.10.3, 4.10.4, 4.10.5 and 4.10.6".

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4.6, Line 14: Delete "Each unit shall also be monitored to determine if intermittencies exist." and substitute "Each unit shall also be monitored for at least one (1) minute during each measurement to determine if intermittencies exist."

4.6, Line 15: Delete "An intermittency of one millisecond or greater duration will be considered to be a failure." and substitute "An intermittency will be considered to be a failure."

Page 2

4.8.1 thru 4.8.2.1: Delete in its entirety and substitute "4.8.1 Temperature-Altitude Comb. - The unit(s) shall meet the operating, non-operating tests of MIL-T-5422 over the altitude and temperature range at supply voltage from 24 to 29 VDC. These tests shall be made for compliance with 3.14.1."

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4.8.12.2, Line 6: Delete "1 C/minute" and substitute "1°C/minute".

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4.9.2.12.3: Delete "with input level set at 1000 to 100,000 microvolts" and substitute "with input level set at 1000 to 10,000 microvolts".

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4.9.3.4: Delete in its entirety and substitute "4.9.3.4 Switch, reversing, loop. - Place the loop reversing switch to top position, select the loop mode of operation, operate the loop L/R switch to position "R", the bearing indicator shall rotate counter-clockwise. Operate the loop L/R switch to position "L", the bearing indicator shall rotate clockwise. Place the loop reversing switch to the bottom position, select the loop mode of operation, operate the loop L/R switch to position "R", the bearing indicator shall rotate clockwise: operate the loop L/R switch to position "L", the bearing indicator shall rotate counter-clockwise. These switch positions shall be tested to determine compliance with 3.15.3.4."

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4.9.4.14: Delete in its entirety and substitute "4.9.4.14 Filtering, Sense and Loop Preamplifiers. - Set RF to 30 MHz, nominally 50,000 microvolts per meter (see para.4.7h) unmodulated, and determine that linear operation of the preamplifiers is maintained. Measure the output of the loop preamplifiers and sense preamplifiers, at the respective test points on the goniometer and receiver combiners, of both an unfiltered reference set of amplifiers and filtered amplifiers under test, assuring that a measurable signal (min 6db S+N/N) at 26 and 34 db below reference is obtained at the output of the filtered amplifiers. Observe the difference between the filtered and unfiltered amplifiers. Repeat RF frequencies of 40, 50, 60, 70, 76, 115, 125, 135, 145, 225, 250, 300 and 400 MHz. Set the RF generator to 100 KHz, 1000 microvolts per meter (see paragraph 4.7h) unmodulated and measure the phase difference between the loop and sense signals at the receiver combiner stages. Repeat for RF frequencies of 200, 400, 1000, 2000 and 3000 KHz. The measurements shall be taken to determine compliance with 3.15.4.14."

4.9.6: Delete in its entirety and substitute "4.9.6 Maximum RF Input. - With the unit operating, apply 1 MHz signal to the input and measure the input level. Remove the 1 MHz signal, apply 50 volts dc to the input for one (1) second. Remove the 50 volts dc and short the input to ground; repeat five (5) times. Apply 100 volts RF signal at 1 MHz through a 50 picofarad capacitor to the input for one (1) minute. Measure the sensitivity of the unit after each voltage application for compliance with 3.15.6."

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Under 4.10.2, insert the following:

"4.10.3 Method 3001 and 3002, Conducted Interference (30 Hz to 100 MHz). - The test shall be performed in accordance with Methods 3001 and 3002 for compliance with 3.16.3.

4.10.4 Method 5005, Rejection of Undesired Signal at Input Terminals (30 Hz to 10 GHz). - The test shall be performed in accordance with Method 5005 for compliance with 3.16.4."

4.10.5 Method 6001, Radiated Susceptibility (14 KHz to 10 GHz). - The test shall be performed in accordance with Method 6001 for compliance with 3.16.5.

4.10.6 Method 5002, Conducted Susceptibility (150 KHz to 400 MHz). - The test shall be performed in accordance with Method 5002 for compliance with 3.16.6."

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5.1.1.4, Line 2: Delete "Figure 1" and substitute "Figure 2".

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Delete "Figure 1" and substitute "Figure 2".

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